

Database Schema Specification

PostgreSQL · DAT-01

1. users

Column	Type	Constraints	Notes
id	UUID	PK , DEFAULT gen_random_uuid()	
email	VARCHAR(255)	NOT NULL, UNIQUE	login identifier
full_name	VARCHAR(120)	NOT NULL	
password_hash	VARCHAR(255)	NOT NULL	bcrypt; omit if Keycloak is the only IdP
role	ENUM user_role	NOT NULL	values: AP_Clerk, Plant_Manager, Finance_Director, VP_Finance
plant_id	VARCHAR(50)	nullable	used for Plant_Manager routing
manager_id	UUID	nullable, FK → users.id	reporting line
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
deleted_at	TIMESTAMPTZ	nullable	soft-delete (paranoid)

Indexes: email (unique), role, plant_id, manager_id.

2. invoices

Column	Type	Constraints	Notes
id	UUID	PK , DEFAULT gen_random_uuid()	
invoice_number	VARCHAR(100)	NOT NULL	
supplier_name	VARCHAR(255)	NOT NULL	
supplier_id	VARCHAR(50)	nullable	
po_number	VARCHAR(60)	nullable	
total_amount	NUMERIC(14,2)	NOT NULL	
currency	CHAR(3)	NOT NULL DEFAULT 'USD'	ISO-4217

Column	Type	Constraints	Notes
status	ENUM invoice_status	NOT NULL DEFAULT 'RECEIVED'	values: RECEIVED, OCR_PROCESSING, PENDING_REVIEW, PENDING_MATCH, MATCHED, PENDING_APPROVAL, APPROVED, REJECTED, EXCEPTION
cfdi_valid	BOOLEAN	nullable	INT-04 gate to MATCHED
ingestion_channel	VARCHAR(50)	nullable	e.g. EMAIL, PORTAL, EDI
plant_id	VARCHAR(50)	nullable	for routing
current_approver_id	UUID	nullable, FK → users.id	
approval_chain	JSONB	nullable	["userId1","userId2",...] (ordered)
approvals_completed	JSONB	nullable	[{approverId, decision, timestamp, notes?}, ...]
rejection_reason	VARCHAR(500)	nullable	
last_escalated_at	TIMESTAMPTZ	nullable	SLA cron (WF-05)
pending_approval_since	TIMESTAMPTZ	nullable	SLA windowing
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
deleted_at	TIMESTAMPTZ	nullable	soft-delete

Indexes: status, current_approver_id, plant_id, (status, pending_approval_since) composite for the WF-05 SLA query.

3. approval_rules

Column	Type	Constraints	Notes
id	UUID	PK , DEFAULT gen_random_uuid()	
rule_name	VARCHAR(60)	NOT NULL, UNIQUE	
min_amount	NUMERIC(14,2)	nullable	inclusive lower bound; NULL = no floor
max_amount	NUMERIC(14,2)	nullable	inclusive upper bound; NULL = catch-all
role_chain	JSONB	NOT NULL	ordered array of user_role values
priority	INTEGER	NOT NULL DEFAULT 100	lower number = checked first
is_active	BOOLEAN	NOT NULL DEFAULT TRUE	
description	VARCHAR(255)	nullable	

Column	Type	Constraints	Notes
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	

Indexes: priority, is_active.

4. audit_logs (append-only)

Column	Type	Constraints	Notes
id	UUID	PK , DEFAULT gen_random_uuid()	
action_type	VARCHAR(60)	NOT NULL	e.g. INVOICE_APPROVED, STATUS_CHANGED
invoice_id	UUID	nullable, FK → invoices.id	
performed_by	UUID	nullable, FK → users.id	
old_value	JSONB	nullable	
new_value	JSONB	nullable	
notes	VARCHAR(500)	nullable	
created_at	TIMESTAMPTZ	NOT NULL DEFAULT NOW()	no updated_at

Special enforcement (DAT-04): trigger that raises an exception on UPDATE or DELETE against this table.

Indexes: invoice_id, performed_by, created_at, action_type.

5. Postgres ENUM types

```
CREATE TYPE user_role AS ENUM (
  'AP_Clerk',
  'Plant_Manager',
  'Finance_Director',
  'VP_Finance'
);

CREATE TYPE invoice_status AS ENUM (
  'RECEIVED',
  'OCR_PROCESSING',
  'PENDING_REVIEW',
  'PENDING_MATCH',
  'MATCHED',
  'PENDING_APPROVAL',
  'APPROVED',
  'REJECTED',
  'EXCEPTION'
);
```

What to ask Roshni when you share this

- Naming convention** — Confirm snake_case column names (Sequelize is configured with underscored: true, so this is what the service expects).
- Soft-delete columns** — users and invoices need deleted_at (paranoid mode). approval_rules and audit_logs don't.

3. **FK behaviour** — invoices.current_approver_id and audit_logs.performed_by should be ON DELETE SET NULL (we don't want a deleted user to nuke history).

4. **Audit-log trigger** — Please own this at the DB level; service code already avoids UPDATE/DELETE but a hard guarantee belongs in the DB.

5. **Vendor / PO master tables** — If she has separate vendors and purchase_orders tables, then invoices.supplier_id and invoices.po_number should become proper FKs instead of free-text.

6. **Plant master** — Same question for plant_id (currently a string; could be FK to a plants table).

Once she shares back her DDL, we line it up against the above and reconcile.